

Renato Mascarenhas Pereira

☎ (+351) 915 142 643 | ✉ renatomascarenhas.pereira@gmail.com | 🏠 renatompereira.pt | 🌐 renatompereira

Experience

Orthopedic Templates and Python Developer

Feb. 2022 - Jan. 2025

PEEKMED

- Prepared and processed 2D/3D CAD orthopedic templates (QCAD, megaCAD, MeshLab) and authored test cases, requirements, and detailed QA protocols for preoperative implant planning (verification & validation).
- Orchestrated full template lifecycle into development/staging/production, and validation, troubleshooting server-side errors and coordinating with the Servers team.
- Engineered and maintained Python scripts and Web apps to automate end-to-end orthopedic template workflows, providing ongoing technical support and reducing manual intervention.
- Managed version control with Bitbucket, tracked issues in JIRA, optimized database performance in MongoDB, and streamlined processes via VS Code and Terminal.

Customer Training

Sept. 2021 – Feb. 2022

ALERT LIFE SCIENCES COMPUTING

- Scheduled and managed internal and customer training sessions, maintained training databases, and generated detailed reports.
- Utilized official government e-platforms to issue and validate training certifications for accredited trainees.
- Designed and executed software test cases, performed functional QA of medical software, logged defects, and worked with developers to ensure quality.
- Prepared and delivered presentations to optimize learning outcomes.

Laboratory technician - Coordinator

May 2021 – Aug. 2022

PRO HEALTH MEDICAL

- Coordinated laboratory operations, optimizing workflows, managing and scheduling multidisciplinary teams, overseeing inventory control, and ensuring timely instrument maintenance.
- Performed and validated both RT-qPCR and LAMP assays, conducted RNA/DNA extraction workflows, and executed comprehensive data analysis to deliver accurate, high-throughput COVID-19 test results.

Laboratory technician

May 2021 – Aug. 2022

UNILABS PORTUGAL

- Performed and validated RT-qPCR assays, conducted RNA/DNA extraction workflows, and executed comprehensive data analysis to deliver accurate, high-throughput COVID-19 test results.

Education

Post-graduation in Clinical Trials

2020 - 2021

FACULTY OF MEDICINE, UNIVERSITY OF LISBON

Masters in Biomedical Engineering

2018 - 2020

FACULTY OF ENGINEERING OF THE UNIVERSITY OF PORTO

Bachelors in Bioengineering

2015 - 2018

FACULTY OF BIOTECHNOLOGY, CATHOLIC UNIVERSITY OF PORTUGAL

Research/Projects

i3S

Sept. 2017 – Oct. 2020

RESEARCH TRAINEE / MASTER'S STUDENT FINALIST

- Working in a research laboratory, BioEngineered Surfaces group, namely focusing on microbiological skills (BSL 2) and development of nanostructured surfaces for bacterial assays, host interactions and cancer.

CBQF

June 2018 – Mar. 2020

PROJECT LEADER / RESEARCH TRAINEE

- Responsible for algorithm/software development for detection of cardiovascular diseases using Machine Learning techniques and medical device development to final product.

Publications

- 2025 **Cholesterol Functionalized Nanoparticles Are Effective against *Helicobacter pylori*, the Gastric Bug: A Proof-of-Concept Study,**
R. Pereira, A. S. Pinho, M. Pereira, A. Rai, L. Ferreira, M. C. L. Martins, P. Parreira
- 2020 **Heart disease detection using ECG lead I and multiple pattern recognition classifiers,**
Pereira, R., Bispo, B., Rodrigues, P. M.

Continuous Education

- 2025 **OrthoMesh3D. Plan orthopaedic surgery in 3D using Blender,** Udemy 9.5h
- 2024 **Building and Deploying an AI-Powered Medical Imaging Chatbot,** Udemy 1.5h
- 2023 **Complete Python Developer in 2023 : Zero to Mastery,** Udemy 31h
- 2021 **3D for Medicine and Orthopedic Surgery,** Udemy 5h
- 2019 **Artificial Intelligence (AI) - Machine Learning for automatization and model prediction,** ESB-UCP 12h

Skills

Languages	Portuguese (native), English (proficient), Spanish (basic)
Management	Agile, JIRA, Project Planning, Team Leadership, Communication
Coding	Python (Pandas, Dash, VTK), Matlab, Arduino, Git, Medical Image
3D Modeling & Printing	SolidWorks, Shapr3D, InVesalius, OsiriX, Rhino, MeshMixer, UltiMaker Cura
Laboratory Techniques	Microbiological & Cellular cultures, Characterization, Analytical Chemistry, Medical instrumentation

Honor and Awards

- 2021 **Merit Award, Scholarship,** Direção-Geral do Ensino Superior, DGES
- 2015 **Merit Award, Scholarship,** Novo Banco